



**Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
*Institut Teknologi Sepuluh Nopember***

Laporan Akhir Praktikum Jaringan Komputer

Vlan-Trunk-OSPF-MultiVendor

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1 Langkah-Langkah Percobaan

A Percobaan 1: Konfigurasi Cisco Switch

(a) Membuat dan mengaktifkan VLAN.

```
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 30
Switch(config-vlan)#
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 40
Switch(config-vlan)#
Switch(config-vlan)#name MARKETING
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#exit
Switch#
Switch#show
*Jun  5 02:16:22.946: N5YS-5-CONFIG_I: Configured from console by console
VLAN Name                Status              Ports
-----
1    default                active              Gi0/0, Gi0/1, Gi0/2, Gi0/3
10   FINANCE                 active              Gi1/0, Gi1/1, Gi1/2, Gi1/3
20   HR                      active
30   IT                      active
40   MARKETING               active
1002 fddi-default            act/unsup
1003 token-ring-default     act/unsup
1004 fddinet-default       act/unsup
1005 trnet-default         act/unsup
Switch#
Switch#
```

(b) Melakukan konfigurasi trunk.

```
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#
Switch(config-if)#description TRUNK-TO-CISCO-ROUTER
Switch(config-if)#switchport mode trunk
Switch(config-if)#
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40
Switch(config-if)#
Switch(config-if)#no shutdown
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#
Switch(config)#exit
Switch#
Switch#show
*Jun  5 02:22:07.368: N5YS-5-CONFIG_I: Configured from console by console
Switch#
Switch#show interfaces trunk
Port      Mode          Encapsulation  Status        Native vlan
Gi0/0     on            802.1q         trunking      1
Gi0/0     Vlans allowed on trunk
          10,20,30,40
Gi0/0     Vlans allowed and active in management domain
          10,20,30,40
Gi0/0     Vlans in spanning tree forwarding state and not pruned
          10,20,30,40
Switch#
Switch#
```

(c) Menyimpan semua konfigurasi yang ada dengan command `write memory`.

```
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40
Switch(config-if)#
Switch(config-if)#no shutdown
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#
Switch(config)#exit
Switch#
Switch#show
*Jun  5 02:22:07.368: N5YS-5-CONFIG_I: Configured from console by console
Switch#
Switch#show interfaces trunk
Port      Mode          Encapsulation  Status        Native vlan
Gi0/0     on            802.1q         trunking      1
Gi0/0     Vlans allowed on trunk
          10,20,30,40
Gi0/0     Vlans allowed and active in management domain
          10,20,30,40
Gi0/0     Vlans in spanning tree forwarding state and not pruned
          10,20,30,40
Switch#
Switch#write memory
Building configuration...
Compressed configuration from 3364 bytes to 1651 bytes[OK]
Switch#
Switch#
*Jun  5 02:22:37.439: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Jun  5 02:22:38.897: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
```

B Percobaan 2: Konfigurasi Cisco Router

(a) Melakukan konfigurasi interface trunk di Switch, subinterface VLAN, serta DHCP-Server untuk VLAN 10 dan 20

```

Jun  5 02:31:51.489: MGRUB-5-CONFIG-WRITTEN: GRUB configuration was written to disk successfully.
Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#exit
Router#
Jun  5 02:32:01.014: NSYS-5-CONFIG-I: Configured from console by console
Router#
Router#write memory
Building configuration...
[OK]
Router#
Jun  5 02:32:12.963: MGRUB-5-CONFIG-WRITING: GRUB configuration is being updated on disk. Please wait...
Jun  5 02:32:13.649: MGRUB-5-CONFIG-WRITTEN: GRUB configuration was written to disk successfully.
Router#
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0      unassigned      YES unset   administratively down down
GigabitEthernet0/1      unassigned      YES unset   administratively down down
GigabitEthernet0/2      unassigned      YES unset   up          up
GigabitEthernet0/2.10   192.168.10.1    YES manual  up          up
GigabitEthernet0/2.20   192.168.20.1    YES manual  up          up
GigabitEthernet0/2.30   192.168.30.1    YES manual  up          up
GigabitEthernet0/2.40   192.168.40.1    YES manual  up          up
GigabitEthernet0/3      unassigned      YES unset   administratively down down
Router#
Router#

```

Gambar 1: Hasil pengecekan show ip interface brief

```

Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0      unassigned      YES unset   administratively down down
GigabitEthernet0/1      unassigned      YES unset   administratively down down
GigabitEthernet0/2      unassigned      YES unset   up          up
GigabitEthernet0/2.10   192.168.10.1    YES manual  up          up
GigabitEthernet0/2.20   192.168.20.1    YES manual  up          up
GigabitEthernet0/2.30   192.168.30.1    YES manual  up          up
GigabitEthernet0/2.40   192.168.40.1    YES manual  up          up
GigabitEthernet0/3      unassigned      YES unset   administratively down down
Router#
Router#show ip dhcp pool

Pool VLAN10-FINANCE :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index   IP address range      Leased addresses
    192.168.10.12    192.168.10.1 - 192.168.10.254 1

Pool VLAN20-HR :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index   IP address range      Leased addresses
    192.168.20.12    192.168.20.1 - 192.168.20.254 1
Router#
Router#

```

Gambar 2: Hasil pengecekan show ip dhcp pool

```

GigabitEthernet0/3      unassigned      YES unset   administratively down down
Router#
Router#show ip dhcp pool

Pool VLAN10-FINANCE :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index   IP address range      Leased addresses
    192.168.10.12    192.168.10.1 - 192.168.10.254 1

Pool VLAN20-HR :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index   IP address range      Leased addresses
    192.168.20.12    192.168.20.1 - 192.168.20.254 1
Router#
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration   Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668-41 Jun 06 2026 02:35 AM Automatic
192.168.20.11    0100.5079.6668-40 Jun 06 2026 02:34 AM Automatic
Router#
Router#

```

Gambar 3: Hasil pengecekan show ip dhcp binding

- (b) Menambahkan konfigurasi IP Static pada PC VLAN 30 dan 40 serta menambahkan konfigurasi `ip dhcp` pada PC VLAN 10 dan 20. Lalu, melakukan pengujian dari PC.

```

0:VPCS
Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Dailing
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirmsh@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" Licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS>
VPCS> ip dhcp
DHCP: IP 192.168.10.11/24 GW 192.168.10.1

VPCS> show ip

NAME      : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER: 192.168.10.1
DHCP LEASE : 86142, 86400/43200/75600
MAC        : 08:50:79:66:68:41
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS>
```

Gambar 4: Memastikan PC VLAN 10 sudah mendapatkan ip dengan command `show ip`

```

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Press '?' to get help.

VPCS>
VPCS> ip dhcp
DHCP: IP 192.168.10.11/24 GW 192.168.10.1

VPCS> show ip

NAME      : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER: 192.168.10.1
DHCP LEASE : 86142, 86400/43200/75600
MAC        : 08:50:79:66:68:41
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.10.1

64 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=2.770 ms
64 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=1.696 ms
64 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=1.437 ms
64 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=1.834 ms
64 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=2.144 ms

VPCS>
```

Gambar 5: Melakukan ping 192.168.10.1 dari PC VLAN 10

```

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Press '?' to get help.

VPCS>
VPCS> ip dhcp
DHCP: IP 192.168.20.11/24 GW 192.168.20.1

VPCS> show ip

NAME      : VPCS[1]
IP/MASK    : 192.168.20.11/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER: 192.168.20.1
DHCP LEASE : 86090, 86400/43200/75600
MAC        : 08:50:79:66:68:40
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.20.1

64 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=1.519 ms
64 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=1.774 ms
64 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=1.447 ms
64 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=1.513 ms
64 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=1.712 ms

VPCS>
```

Gambar 6: Memastikan PC VLAN 20 sudah mendapatkan ip dengan command `show ip` serta mampu melakukan ping 192.168.20.1

```

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Press '?' to get help.

VPCS>
VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PC1 - 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VPCS> show ip

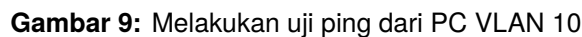
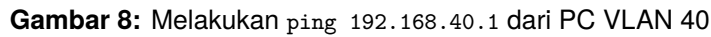
NAME      : VPCS[1]
IP/MASK    : 192.168.30.10/24
GATEWAY    : 192.168.30.1
DNS        :
MAC        : 08:50:79:66:68:3f
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.30.1

64 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=1.825 ms
64 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=1.485 ms
64 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=1.809 ms
64 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=1.746 ms
64 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=1.979 ms

VPCS>
```

Gambar 7: Melakukan ping 192.168.30.1 dari PC VLAN 30



- ```

[00000000] Server: W:4.0.0.250
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool:
Current index : 254
Leased addresses : 192.168.10.12 - 192.168.10.254
Leased addresses : 1

Pool VLAN200-HB :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool:
Current index : 254
Leased addresses : 192.168.20.12 - 192.168.20.254
Leased addresses : 1

Routers
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address Hardware address/
User name
192.168.10.11 0100.5979.6668.41 Jun 06 2026 02:35 AM Automatic
192.168.20.11 0100.5979.6668.40 Jun 06 2026 02:34 AM Automatic
Routers
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address Client-ID/
Hardware address/
User name
192.168.10.11 0100.5979.6668.41 Jun 06 2026 02:35 AM Automatic
192.168.20.11 0100.5979.6668.40 Jun 06 2026 02:34 AM Automatic
Routers

```

(a) Melakukan konfigurasi LAN Huawei dan IP di VPC LAN Huawei lalu melakukan uji ping gateway Huawei.

```

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For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS>
VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address:
PCI : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip
NAME : VPCS[1]
IP/MASK : 192.168.50.10/24
GATEWAY : 192.168.50.1
DNS :
MAC : 08:50:79:66:68:42
LPORT : 20000
RHOST-PORT : 127.6.0.1:30000
MTU : 1500

VPCS> ping 192.168.50.1
84 bytes from 192.168.50.1: icmp_seq=1 ttl=255 time=0.238 ms
84 bytes from 192.168.50.1: icmp_seq=2 ttl=255 time=0.597 ms
84 bytes from 192.168.50.1: icmp_seq=3 ttl=255 time=1.747 ms
84 bytes from 192.168.50.1: icmp_seq=4 ttl=255 time=1.575 ms
84 bytes from 192.168.50.1: icmp_seq=5 ttl=255 time=1.640 ms

VPCS>

```

(b) Melakukan konfigurasi IP link Cisco ke Huawei Link 1, Huawei Link 2, Mikrotik serta konfigurasi IP link Huawei ke Mikrotik. Lalu, melakukan verifikasi interface.

```

Router(config-if)#ip address 10.10.30.1 255.255.255.252
Router(config-if)#
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#exit
Router#
Router#
*Jun 5 02:56:22.471: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
*Jun 5 02:56:22.486: %SYS-5-CONFIG_I: Configured from console by console
*Jun 5 02:56:23.471: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router#
Router#write memory
Building configuration...
[OK]
Router#
Router#
*Jun 5 02:56:29.155: %GRUB-5-CONFIG-WRITTEN: GRUB configuration is being updated on disk. Please wait...
*Jun 5 02:56:29.843: %GRUB-5-CONFIG-WRITTEN: GRUB configuration was written to disk successfully.
Router#
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.10.11.1 YES manual up up
GigabitEthernet0/1 10.10.30.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset up up
GigabitEthernet0/2.10 192.168.10.1 YES manual up up
GigabitEthernet0/2.20 192.168.20.1 YES manual up up
GigabitEthernet0/2.30 192.168.30.1 YES manual up up
GigabitEthernet0/2.40 192.168.40.1 YES manual up up
GigabitEthernet0/3 10.10.10.1 YES manual up up
Router#
Router#

```

Gambar 10: Tampilan Cisco Router

```

[~HUAWEI-Ethernet1/0/3]
[~HUAWEI-Ethernet1/0/3]undo shutdown
[~HUAWEI-Ethernet1/0/3]
[~HUAWEI-Ethernet1/0/3]quit
[~HUAWEI]
[~HUAWEI]commit
[~HUAWEI]
[~HUAWEI]display ip interface brief
Idom: administratively down
Idom: FIB overload down
Idom: standby
(L): loopback
(C): spoofing
(D): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface IP Address/Mask Physical Protocol VPN
Ethernet1/0/0 10.10.20.2/30 up up l3vpn
Ethernet1/0/1 unassigned up down --
Ethernet1/0/2 128.1.138.137/16 up up l3vpn
Ethernet1/0/3 192.168.50.1/24 up up --
Ethernet1/0/4 128.1.138.137/16 up up l3vpn
Ethernet1/0/5 10.10.20.1/30 up up --
Ethernet1/0/6 128.1.138.137/16 up up l3vpn
Ethernet1/0/7 10.10.21.2/30 up up --
Ethernet1/0/8 128.1.138.137/16 up up l3vpn
Ethernet1/0/9 unassigned up down --
----- More -----

```

Gambar 11: Tampilan Huawei Router

```

MikroTik RouterOS 6.49.17 (c) 1999-2024 http://www.mikrotik.com/

Do you want to see the software license? [y/n]: n
[?] Gives the list of available commands
command [?] Gives help on the command and list of arguments

[Tab] Completes the command/word. If the input is ambiguous,
a second [Tab] gives possible options

/ Move up to base level
./ Move up one level
/command Use command at the base level
Jun/05/2025 02:13:59 system,error,critical router was rebooted without proper shutdown

Change your password
new password>

[admin@Mikrotik] > /ip address add address=10.10.20.2/30 interface=ether1 comment=TO-HUAWEI
[admin@Mikrotik] > /ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CISCO
[admin@Mikrotik] > /ip address print
Flags: K - disabled, I - invalid, D - dynamic
ADDRESS NETWORK INTERFACE
0 ::: TO-HUAWEI 10.10.20.0 ether1
1 ::: TO-CISCO 10.10.30.0 ether2
[admin@Mikrotik] >

```

Gambar 12: Tampilan Mikrotik

(c) Melakukan ping antar router.

```

[OK]
Router#
Router#
*Jun 5 02:56:29.155: MGRUB-S-CONFIG-WRITING: GRUB configuration is being updated on disk. Please wait...
*Jun 5 02:56:29.843: MGRUB-S-CONFIG-WRITTEN: GRUB configuration was written to disk successfully.
Router#
Router#show ip interface brief
Interface IP-Address HW Method Status Protocol
GigabitEthernet0/0 10.10.11.1 VES manual up up
GigabitEthernet0/1 10.10.30.1 VES manual up up
GigabitEthernet0/2 unassigned VES unset up up
GigabitEthernet0/2.10 192.168.10.1 VES manual up up
GigabitEthernet0/2.20 192.168.20.1 VES manual up up
GigabitEthernet0/2.30 192.168.30.1 VES manual up up
GigabitEthernet0/2.40 192.168.40.1 VES manual up up
GigabitEthernet0/3 10.10.10.1 VES manual up up
Router#
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 88 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!
Success rate is 88 percent (4/5), round-trip min/avg/max = 1/1/3 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
!!!!
Success rate is 88 percent (4/5), round-trip min/avg/max = 1/1/1 ms
Router#

```

**Gambar 13:** Tampilan Cisco Router

```

[huawei]
Ethernet0/0/0/4094 128.1.138.197/16 up up l3vpn
Ethernet1/0/0 unassigned down --
[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=2 ms
Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=1 ms
Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms
--- 10.10.10.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/2 ms
[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=2 ms
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=1 ms
--- 10.10.11.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/2 ms
[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=2 ms
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=1 ms
--- 10.10.11.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/2 ms
[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms
--- 10.10.20.2 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/1 ms
[~HUAWEI]
[~HUAWEI]

```

**Gambar 14:** Tampilan Huawei Router

```

[Mikrotik]
Flags: X - disabled, I - invalid, D - dynamic
ADDRESS NETWORK INTERFACE
0 :::: TO-HUAWEI 10.10.20.2/30 10.10.20.0 ether1
1 :::: TO-CISCO 10.10.30.2/30 10.10.30.0 ether2
[Admin@Mikrotik] > ping 10.10.20.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.20.1 56 255 1ms
1 10.10.20.1 56 255 0ms
2 10.10.20.1 56 255 0ms
3 10.10.20.1 56 255 0ms
4 10.10.20.1 56 255 0ms
5 10.10.20.1 56 255 0ms
6 10.10.20.1 56 255 0ms
7 10.10.20.1 56 255 0ms
8 10.10.20.1 56 255 0ms
9 10.10.20.1 56 255 0ms
10 10.10.20.1 56 255 0ms
sent=11 received=11 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=1ms
[Admin@Mikrotik] > ping 10.10.30.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.30.1 56 255 0ms
1 10.10.30.1 56 255 1ms
2 10.10.30.1 56 255 1ms
3 10.10.30.1 56 255 0ms
4 10.10.30.1 56 255 1ms
5 10.10.30.1 56 255 2ms
6 10.10.30.1 56 255 0ms
sent=7 received=7 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=2ms
[Admin@Mikrotik] >

```

**Gambar 15:** Tampilan Mikrotik

## D Percobaan 4: Konfigurasi OSPF Multivendor dan OSPF Cost

(a) Melakukan konfigurasi OSPF pada Cisco Router dan pastikan OSPF muncul.

```
Router#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "application"
 Sending updates every 0 seconds
 Invalid after 0 seconds, hold down 0, flushed after 0
 Outgoing update filter list for all interfaces is not set
 Incoming update filter list for all interfaces is not set
 Maximum path: 32
 Routing for Networks:
 Routing Information Sources:
 Gateway Distance Last Update
 Distance: (default is 0)

Routing Protocol is "ospf 1"
 Outgoing update filter list for all interfaces is not set
 Incoming update filter list for all interfaces is not set
 Router ID 1.1.1.1
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 Maximum path: 4
 Routing for Networks:
 10.10.10.0 0.0.0.3 area 0
 10.10.11.0 0.0.0.3 area 0
 10.10.30.0 0.0.0.3 area 0
 192.168.10.0 0.0.0.255 area 0
 192.168.20.0 0.0.0.255 area 0
 192.168.30.0 0.0.0.255 area 0
 192.168.40.0 0.0.0.255 area 0
 Routing Information Sources:
 Gateway Distance Last Update
 Distance: (default is 110)

Router#
```

(b) Melakukan konfigurasi OSPF pada Huawei Router dan verifikasi OSPF Huawei.

```
0.00% packet loss
round-trip min/avg/max = 1/1/5 ms

[~HUAWEI]
[~HUAWEI]display ospf peer
(N) Indicates MADJ neighbor

OSPF Process 1 with Router ID 2.2.2.2
 Neighbors
Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.1 BDR: 10.10.10.2 MTU: 1500
Dead timer due in 37 sec
Retrans timer interval: 5
Neighbor is up for 00:00:00:00
Neighbor Up Time : 2026-06-05 03:20:50
Authentication Sequence: [0]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 36 sec
Retrans timer interval: 5
Neighbor is up for 00:00:00:00
Neighbor Up Time : 2026-06-05 03:20:50
Authentication Sequence: [0]

[~HUAWEI]
```

(c) Memastikan Huawei telah peer dengan Cisco melalui 2 link utama dan sudah mendapatkan network lengkap dari Cisco.

```
public Routing Table : OSPF
 Destinations : 9 Routes : 14
OSPF routing table status : <Active>
 Destinations : 5 Routes : 10
Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.30.0/30 OSPF 10 110 0 10.10.11.1 Ethernet1/0/4
192.168.10.0/24 OSPF 10 110 0 10.10.10.1 Ethernet1/0/0
192.168.20.0/24 OSPF 10 11 0 10.10.10.1 Ethernet1/0/0
192.168.30.0/24 OSPF 10 11 0 10.10.11.1 Ethernet1/0/4
192.168.40.0/24 OSPF 10 11 0 10.10.11.1 Ethernet1/0/4
192.168.50.0/24 OSPF 10 11 0 10.10.11.1 Ethernet1/0/4
192.168.50.0/24 OSPF 10 11 0 10.10.10.1 Ethernet1/0/0

OSPF routing table status : <Inactive>
 Destinations : 4 Routes : 4
Destination/Mask Proto Pre Cost Flags NextHop Interface
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/0
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/4
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/3
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2

[~HUAWEI]
[~HUAWEI]
[~HUAWEI]
[~HUAWEI]

[~HUAWEI]
[~HUAWEI]display ospf routing

OSPF Process 1 with Router ID 2.2.2.2
 Routing Tables

Routing for Network
Destination Cost Type NextHop AdvRouter Area
10.10.10.0/30 10 Direct 10.10.10.2 2.2.2.2 0.0.0.0
10.10.11.0/30 10 Direct 10.10.11.2 2.2.2.2 0.0.0.0
10.10.20.0/30 100 Direct 10.10.20.1 2.2.2.2 0.0.0.0
10.10.30.0/30 110 Stub 10.10.11.1 1.1.1.1 0.0.0.0
10.10.30.0/30 110 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.50.0/24 1 Direct 192.168.50.1 2.2.2.2 0.0.0.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0

---- More ----
```

(d) Melakukan konfigurasi OSPF pada Mikrotik dan melakukan verifikasi OSPF Mikrotik.



```

9 10.10.20.1 56 255 0ms
10 10.10.20.1 56 255 0ms
sent=11 received=11 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=1ms

[admin@MikroTik] > ping 10.10.30.1
PING: 10.10.30.1 56 255 0ms
0 10.10.30.1 56 255 0ms
1 10.10.30.1 56 255 0ms
2 10.10.30.1 56 255 0ms
3 10.10.30.1 56 255 0ms
4 10.10.30.1 56 255 0ms
5 10.10.30.1 56 255 0ms
6 10.10.30.1 56 255 0ms
sent=7 received=7 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=2ms

[admin@MikroTik] > /routing ospf instance set default router-id=3.3.3
[admin@MikroTik] > /routing ospf
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@MikroTik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@MikroTik] > /routing ospf /
[admin@MikroTik] > /routing ospf interface
[admin@MikroTik] > /routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] > /routing ospf interface /
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
 backup-dr-address=10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
 adjacency=125
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
 backup-dr-address=10.10.20.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
 adjacency=125
[admin@MikroTik] >

```

(e) Memastikan router ID Cisco dan Huawei sudah muncul di Mikrotik.

```

[admin@MikroTik] > /routing ospf instance set default router-id=3.3.3
[admin@MikroTik] > /routing ospf
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@MikroTik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@MikroTik] > /routing ospf /
[admin@MikroTik] > /routing ospf interface
[admin@MikroTik] > /routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
 backup-dr-address=10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
 adjacency=125
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
 backup-dr-address=10.10.20.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
 adjacency=125
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, n - mme,
a - blackhole, U - unreachable, P - prohibit
DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 Ado 10.10.10.0/30 10.10.20.1 110
1 Ado 10.10.11.0/30 10.10.30.1 110
2 Ado 192.168.10.0/24 10.10.30.1 110
3 Ado 192.168.20.0/24 10.10.30.1 110
4 Ado 192.168.30.0/24 10.10.30.1 110
5 Ado 192.168.40.0/24 10.10.30.1 110
6 Ado 192.168.50.0/24 10.10.30.1 110
[admin@MikroTik] >

```

(f) Melakukan verifikasi OSPF neighbor di masing-masing router.

```

Distance: (default is 0)

Routing Protocol is "ospf 1"
 Outgoing update filter list for all interfaces is not set
 Incoming update filter list for all interfaces is not set
 Router ID 1.1.1.1
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 Maximum path: 4
 Routing for Networks:
 10.10.10.0 0.0.0.3 area 0
 10.10.11.0 0.0.0.3 area 0
 10.10.30.0 0.0.0.3 area 0
 192.168.10.0 0.0.0.255 area 0
 192.168.20.0 0.0.0.255 area 0
 192.168.30.0 0.0.0.255 area 0
 192.168.40.0 0.0.0.255 area 0
 192.168.50.0 0.0.0.255 area 0
 Routing Information Sources:
 Gateway Distance
 Distance: (default is 110)

Router#
*Jun 5 03:20:38.246: %OSPF-3-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from LOADING to FULL, Loading Done
*Jun 5 03:20:40.097: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from LOADING to FULL, Loading Done
Router#
*Jun 5 03:27:38.478: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
*Jun 5 03:28:01.463: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
Router#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
1.3.3.3 1 FULL/DR 00:00:36 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/DR 00:00:36 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/DR 00:00:29 10.10.10.2 GigabitEthernet0/3
Router#

```

Gambar 16: Tampilan Cisco Router

```

Neighbors
Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:DR is Slave Priority: 1
DR: 10.10.10.1 BDR: 10.10.10.2 MTU: 1500
Dead timer due in 36 sec
Neighbor is up for 00h00m34s
Retrans timer interval: 5
Authentication Sequence: [0]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3 Address: 10.10.20.2
State: Full Mode:DR is Master Priority: 1
DR: 10.10.20.1 BDR: 10.10.20.2 MTU: 1500
Dead timer due in 37 sec
Retrans timer interval: 5
Neighbor is up for 00h01m11s
Neighbor Up Time : 2026-06-05 03:20:58
Authentication Sequence: [0]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:DR is Slave Priority: 1
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 36 sec
Neighbor is up for 00h00m27s
Neighbor Up Time : 2026-06-05 03:20:58
Authentication Sequence: [0]

[~HUAWEI]

```

Gambar 17: Tampilan Huawei Router

**Gambar 18:** Tampilan Mikrotik

```

10.10.10.0 0.0.0.3 area 0
10.10.10.0 0.0.0.2 area 0
10.10.10.0 0.0.0.3 area 0
10.10.10.0 0.0.0.2 area 0
102.168.18.0 0.0.0.255 area 0
102.168.20.0 0.0.0.255 area 0
102.168.38.0 0.0.0.255 area 0
102.168.40.0 0.0.0.255 area 0

Routing Information Sources:
Gateway Distance Last Update
Distance: (default is 140)

Routers
Router# 5 03:20:38.206: OSPF>F-A-DJCHG: Process 1, Mbr 2.2.2.2 on GigabitEthernet0/3 from LOADING to FULL, Loading Done
Router# 5 03:20:40.569: OSPF>F-A-DJCHG: Process 1, Mbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
Router# 5 03:27:30.708: OSPF>F-A-DJCHG: Process 1, Mbr 3.3.3.3 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
Router# 5 03:28:01.463: OSPF>F-A-DJCHG: Process 1, Mbr 3.3.3.3 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
Router#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
2.2.2.2 1 FULL/DR 00:00:38 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/DR 00:00:36 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/DR 00:01:29 10.10.10.2 GigabitEthernet0/3
Router#show ip route 102.168.0.0
Routing entry for 102.168.0.0/24
 Summary written to interface
 Nbrs in "ospf 1", distance 118, metric 1, type intra area
 Last update from 10.18.11.2 on GigabitEthernet0/0, 00:09:34 ago
 Routing Descriptor Blocks:
 10.10.11.2, from 2.2.2.2, 00:09:34 ago, via GigabitEthernet0/0
 Route metric is 11, traffic share count is 1
 10.10.10.2, from 2.2.2.2, 00:09:44 ago, via GigabitEthernet0/3
 Route metric is 11, traffic share count is 1
 10.10.10.2
Router#

```

**Gambar 19:** Tampilan Cisco Router

**Gambar 20:** Tampilan Huawei Router

**Gambar 21:** Tampilan Mikrotik

9

```

VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=3.717 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=3.631 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=3.539 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=3.068 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=2.915 ms

VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=1.927 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=3.138 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=2.899 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=3.653 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=3.067 ms

VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=2.632 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=2.809 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=2.148 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=3.838 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=3.395 ms

VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.538 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.852 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.165 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.622 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.386 ms

VPCS>

```

**Gambar 22:** Tampilan PC VLAN 10

```

VPCS> ip dhcp
DHCPA IP 192.168.20.11/24 GW 192.168.20.1

VPCS> show ip

NAME : VPCS[1]
IP/MASK : 192.168.20.11/24
GATEWAY : 192.168.20.1
DNS : 8.8.8.8
DHCP SERVER: 192.168.20.1
DHCP LEASE : 86400, 86400/43200/75600
MAC : 08:50:79:66:68:40
LPORT : 20000
RHOST-PORT : 127.0.0.1:30000
MTU : 1500

VPCS> ping 192.168.20.1
84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=1.539 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=2.974 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=1.417 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=1.513 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=1.712 ms

VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=5.207 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.655 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.313 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.561 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.373 ms

VPCS>

```

**Gambar 23:** Tampilan PC VLAN 20

```

VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PCI : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VPCS> show ip

NAME : VPCS[1]
IP/MASK : 192.168.30.10/24
GATEWAY : 192.168.30.1
DNS :
MAC : 08:50:79:66:68:3f
LPORT : 20000
RHOST-PORT : 127.0.0.1:30000
MTU : 1500

VPCS> ping 192.168.30.1
84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=1.825 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=1.485 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=1.899 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=1.946 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=1.979 ms

VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=4.258 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=2.974 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=2.844 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.813 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.905 ms

VPCS>

```

**Gambar 24:** Tampilan PC VLAN 30

```

VPCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
*HPC1 : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

VPCS> show ip

NAME : VPCS[1]
IP/MASK : 192.168.40.10/24
GATEWAY : 192.168.40.1
DNS :
MAC : 08:50:79:66:68:3e
LPORT : 20000
RHOST-PORT : 127.0.0.1:30000
MTU : 1500

VPCS> ping 192.168.40.1
84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=1.617 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=1.731 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=2.004 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=1.394 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=1.175 ms

VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=2.846 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.865 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.540 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.746 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=4.141 ms

VPCS>

```

**Gambar 25:** Tampilan PC VLAN 40

```
PC-Huawei:~$ ping 192.168.10.11
PING: 64 bytes from 192.168.10.11: icmp_seq=1 ttl=62 time=3.986 ms
PING: 64 bytes from 192.168.10.11: icmp_seq=2 ttl=62 time=5.836 ms
PING: 64 bytes from 192.168.10.11: icmp_seq=3 ttl=62 time=5.975 ms
PING: 64 bytes from 192.168.10.11: icmp_seq=4 ttl=62 time=3.681 ms
PING: 64 bytes from 192.168.10.11: icmp_seq=5 ttl=62 time=4.110 ms

PC-Huawei:~$ ping 192.168.20.11
PING: 64 bytes from 192.168.20.11: icmp_seq=1 ttl=62 time=4.620 ms
PING: 64 bytes from 192.168.20.11: icmp_seq=2 ttl=62 time=9.432 ms
PING: 64 bytes from 192.168.20.11: icmp_seq=3 ttl=62 time=5.534 ms
PING: 64 bytes from 192.168.20.11: icmp_seq=4 ttl=62 time=3.633 ms
PING: 64 bytes from 192.168.20.11: icmp_seq=5 ttl=62 time=3.803 ms

PC-Huawei:~$ ping 192.168.30.10
PING: 64 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=3.231 ms
PING: 64 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=9.432 ms
PING: 64 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=4.670 ms
PING: 64 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=3.897 ms
PING: 64 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=4.630 ms

PC-Huawei:~$ ping 192.168.40.10
PING: 64 bytes from 192.168.40.10: icmp_seq=1 ttl=62 time=4.288 ms
PING: 64 bytes from 192.168.40.10: icmp_seq=2 ttl=62 time=7.892 ms
PING: 64 bytes from 192.168.40.10: icmp_seq=3 ttl=62 time=3.721 ms
PING: 64 bytes from 192.168.40.10: icmp_seq=4 ttl=62 time=6.083 ms
PING: 64 bytes from 192.168.40.10: icmp_seq=5 ttl=62 time=4.352 ms

PC-Huawei:~$
```

**Gambar 26:** Hasil pengujian dari PC LAN Huawei ke IP VLAN Cisco

## E Percobaan 5: Pengujian Failover OSPF

- (a) Pada kondisi normal yaitu 2 link Cisco-Huawei aktif, hasilnya traffic melewati jalur langsung Cisco > Huawei.

```
PC-Cisco:~$ ping 192.168.30.1
PING: 64 bytes from 192.168.30.1: icmp_seq=1 ttl=255 time=1.825 ms
PING: 64 bytes from 192.168.30.1: icmp_seq=2 ttl=255 time=1.485 ms
PING: 64 bytes from 192.168.30.1: icmp_seq=3 ttl=255 time=1.899 ms
PING: 64 bytes from 192.168.30.1: icmp_seq=4 ttl=255 time=1.746 ms
PING: 64 bytes from 192.168.30.1: icmp_seq=5 ttl=255 time=1.979 ms

PC-Cisco:~$ ping 192.168.50.10
PING: 64 bytes from 192.168.50.10: icmp_seq=1 ttl=62 time=4.258 ms
PING: 64 bytes from 192.168.50.10: icmp_seq=2 ttl=62 time=2.974 ms
PING: 64 bytes from 192.168.50.10: icmp_seq=3 ttl=62 time=2.944 ms
PING: 64 bytes from 192.168.50.10: icmp_seq=4 ttl=62 time=3.813 ms
PING: 64 bytes from 192.168.50.10: icmp_seq=5 ttl=62 time=3.905 ms

PC-Cisco:~$ trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.30.1 4.383 ms 2.333 ms 2.456 ms
 2 10.10.10.2 4.367 ms 2.418 ms 3.359 ms
 3 *192.168.50.10 3.744 ms (ICMP type=3, code=3, Destination port unreachable)

PC-Cisco:~$
```

- (b) Ketika kondisi 1 link Cisco-Huawei mati, hasil traffic tetap melewati jalur langsung Cisco.

```
PC-Cisco:~$ ping 192.168.30.1
PING: 64 bytes from 192.168.30.1: icmp_seq=1 ttl=255 time=1.825 ms
PING: 64 bytes from 192.168.30.1: icmp_seq=2 ttl=255 time=1.485 ms
PING: 64 bytes from 192.168.30.1: icmp_seq=3 ttl=255 time=1.899 ms
PING: 64 bytes from 192.168.30.1: icmp_seq=4 ttl=255 time=1.746 ms
PING: 64 bytes from 192.168.30.1: icmp_seq=5 ttl=255 time=1.979 ms

PC-Cisco:~$ ping 192.168.50.10
PING: 64 bytes from 192.168.50.10: icmp_seq=1 ttl=62 time=4.258 ms
PING: 64 bytes from 192.168.50.10: icmp_seq=2 ttl=62 time=2.974 ms
PING: 64 bytes from 192.168.50.10: icmp_seq=3 ttl=62 time=2.944 ms
PING: 64 bytes from 192.168.50.10: icmp_seq=4 ttl=62 time=3.813 ms
PING: 64 bytes from 192.168.50.10: icmp_seq=5 ttl=62 time=3.905 ms

PC-Cisco:~$ trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.30.1 4.383 ms 2.333 ms 2.456 ms
 2 10.10.10.2 4.367 ms 2.418 ms 3.359 ms
 3 *192.168.50.10 3.744 ms (ICMP type=3, code=3, Destination port unreachable)

PC-Cisco:~$ trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.30.1 4.328 ms 1.638 ms 1.372 ms
 2 10.10.11.2 2.720 ms 3.551 ms 2.539 ms
 3 *192.168.50.10 5.579 ms (ICMP type=3, code=3, Destination port unreachable)

PC-Cisco:~$
```

- (c) Ketika 2 link Cisco-Huawei mati, hasilnya rute berpindah ke jalur backup melalui MikroTik.

```

84 bytes from 192.168.38.1: icmp_seq=2 ttl=255 time=1.485 ms
84 bytes from 192.168.38.1: icmp_seq=3 ttl=255 time=1.899 ms
84 bytes from 192.168.38.1: icmp_seq=4 ttl=255 time=1.796 ms
84 bytes from 192.168.38.1: icmp_seq=5 ttl=255 time=1.979 ms

VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10: icmp_seq=1 ttl=62 time=4.258 ms
84 bytes from 192.168.50.10: icmp_seq=2 ttl=62 time=2.974 ms
84 bytes from 192.168.50.10: icmp_seq=3 ttl=62 time=2.904 ms
84 bytes from 192.168.50.10: icmp_seq=4 ttl=62 time=3.813 ms
84 bytes from 192.168.50.10: icmp_seq=5 ttl=62 time=3.905 ms

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.38.1 4.343 ms 2.333 ms 2.456 ms
 2 10.10.10.2 4.367 ms 2.418 ms 3.359 ms
 3 *192.168.50.10 3.704 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.38.1 1.928 ms 1.638 ms 3.372 ms
 2 10.10.11.2 2.720 ms 3.551 ms 2.538 ms
 3 *192.168.50.10 5.579 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.38.1 3.862 ms 2.728 ms 4.975 ms
 2 10.10.38.2 3.423 ms 4.426 ms 5.813 ms
 3 10.10.20.1 6.538 ms 6.405 ms 2.458 ms
 4 *192.168.50.10 3.976 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS>

```

(d) Ketika kedua link diaktifkan kembali, jalur routingnya harus kembali berpindah ke link utama Cisco-Huawei.

```

84 bytes from 192.168.50.10: icmp_seq=1 ttl=62 time=4.258 ms
84 bytes from 192.168.50.10: icmp_seq=2 ttl=62 time=2.974 ms
84 bytes from 192.168.50.10: icmp_seq=3 ttl=62 time=2.904 ms
84 bytes from 192.168.50.10: icmp_seq=4 ttl=62 time=3.813 ms
84 bytes from 192.168.50.10: icmp_seq=5 ttl=62 time=3.905 ms

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.38.1 4.343 ms 2.333 ms 2.456 ms
 2 10.10.10.2 4.367 ms 2.418 ms 3.359 ms
 3 *192.168.50.10 3.704 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.38.1 1.928 ms 1.638 ms 3.372 ms
 2 10.10.11.2 2.720 ms 3.551 ms 2.538 ms
 3 *192.168.50.10 5.579 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.38.1 3.862 ms 2.728 ms 4.975 ms
 2 10.10.38.2 3.423 ms 4.426 ms 5.813 ms
 3 10.10.20.1 6.538 ms 6.405 ms 2.458 ms
 4 *192.168.50.10 3.976 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.38.1 2.482 ms 2.111 ms 2.972 ms
 2 10.10.10.2 2.911 ms 1.596 ms 2.265 ms
 3 *192.168.50.10 3.728 ms (ICMP type=3, code=3, Destination port unreachable)

VPCS>

```

## 2 Analisis Hasil Percobaan

Praktikum jaringan komputer modul 5 yang berjudul VLAN-Trunk-OSPF-MultiVendor memiliki 5 percobaan di dalamnya. Percobaan yang pertama adalah melakukan konfigurasi VLAN dan trunk pada Cisco Switch. Dengan pembuatan VLAN, setiap kelompok pengguna ditempatkan pada domain *broadcast* yang berbeda sehingga lalu lintas jaringan menjadi lebih terorganisir dan aman. Konfigurasi *access port* memastikan setiap PC masuk ke VLAN yang sesuai, sedangkan konfigurasi trunk pada *interface* yang terhubung ke router memungkinkan beberapa VLAN melewati satu *link* fisik menggunakan mekanisme IEEE 802.1Q tagging. Hasil menunjukkan seluruh VLAN aktif serta trunk membawa VLAN 10, 20, 30, dan 40 membuktikan bahwa konfigurasi antar perangkat telah berjalan sesuai teori VLAN dan trunking.

Beralih ke percobaan yang kedua yaitu implementasi Inter-VLAN routing menggunakan metode Router-on-a-Stick (ROAS). Metode ini memungkinkan perangkat dalam VLAN berbeda berkomunikasi secara langsung meski berada pada jaringan logis yang berbeda. Router akan dibuat memiliki beberapa *subinterface* yang masing-masing mewakili *gateway* dari setiap VLAN. Ketika berhasil melakukan ping ke gateway masing-masing dan perangkat antar VLAN dapat saling berkomunikasi, hal tersebut menunjukkan bahwa proses routing sudah benar. DHCP pada VLAN 10 dan VLAN 20 yang berhasil juga menunjukkan bahwa router tidak hanya berfungsi sebagai gateway tetapi juga sebagai DHCP server yang mampu mendistribusikan alamat IP secara otomatis kepada klien.

Percobaan yang ketiga membahas tentang konektivitas antar router multivendor. Seperti yang terlihat di topologi, terdapat 3 jenis router, yaitu Cisco, Huawei, dan MikroTik. Ketiga router tersebut harus

bisa berkomunikasi menggunakan standar protokol jaringan yang sama. Komunikasi antar perangkat jaringan tidak bergantung pada vendor yang digunakan, melainkan pada kesesuaian implementasi standar protokol TCP/IP. Hasil pengujian yang berhasil membuktikan bahwa proses *addressing*, *sub-netting*, dan konfigurasi *interface* telah dilakukan dengan benar.

Selanjutnya adalah percobaan keempat yang membuktikan implementasi OSPF pada lingkungan multivendor. Secara teori, OSPF merupakan protokol routing dinamis berbasis *link-state* yang mampu bertukar informasi routing antar perangkat secara otomatis melalui pembentukan hubungan *neighbor*. Ketika Cisco berhasil membentuk *neighbor* dengan Huawei dan MikroTik, Huawei berhasil membentuk *peer* dengan Cisco dan MikroTik, serta seluruh router memperoleh route menuju seluruh network yang ada, maka dapat disimpulkan bahwa proses pertukaran *Link State Advertisement* (LSA) berlangsung dengan baik. Selain itu, keberhasilan komunikasi antara PC di VLAN Cisco dengan LAN Huawei menunjukkan bahwa seluruh jaringan telah dipelajari dan dimasukkan ke *routing table* masing-masing router. Hasil ini membuktikan bahwa standar OSPF dapat bekerja dengan baik pada perangkat Cisco, Huawei, dan MikroTik tanpa memerlukan protokol khusus dari vendor tertentu.

Percobaan kelima menunjukkan salah satu keunggulan utama OSPF, yaitu kemampuan memilih jalur terbaik secara otomatis berdasarkan nilai *cost* dan melakukan *failover* ketika terjadi gangguan jaringan. Pada kondisi normal, lalu lintas data menuju jaringan Huawei melewati dua *link* utama Cisco-Huawei karena memiliki *cost* yang lebih rendah. Ketika salah satu *link* dimatikan, OSPF tetap mempertahankan konektivitas dengan menggunakan *link* utama yang masih aktif. Saat kedua *link* utama dimatikan, OSPF secara otomatis menghitung ulang topologi jaringan dan mengalihkan trafik melalui jalur cadangan Cisco > MikroTik > Huawei yang memiliki *cost* lebih tinggi. Setelah kedua *link* utama diaktifkan kembali, OSPF kembali memilih jalur utama karena memiliki metrik terbaik. Hasil ini sesuai dengan teori algoritma *Shortest Path First* (SPF) yang digunakan OSPF untuk menentukan rute paling efisien sekaligus menyediakan redundansi jaringan tanpa memerlukan konfigurasi ulang secara manual.

### 3 Hasil Tugas Modul

Hasil pengerjaan tugas modul dapat dilihat di README.md folder Modul 5/LA di GitHub Repository Kelompok 23

### 4 Kesimpulan

Berdasarkan praktikum modul 5 jaringan komputer yang telah dilakukan, dapat disimpulkan bahwa VLAN bisa memisahkan jaringan secara logis, trunk memungkinkan banyak VLAN melewati satu media fisik, Router-on-a-Stick memungkinkan komunikasi antar VLAN, DHCP mempermudah pengelolaan alamat IP, sedangkan OSPF memungkinkan pertukaran informasi routing secara otomatis antar perangkat Cisco, Huawei, dan MikroTik. Pengujian failover yang berhasil juga membuktikan bahwa jaringan memiliki tingkat ketersediaan yang lebih tinggi karena mampu beradaptasi terhadap kegagalan *link* tanpa mengganggu komunikasi antar jaringan.

## 5 Lampiran

### 5.1 Dokumentasi saat praktikum

